

Lesson 14 · Definite integrals, area and accumulation

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q1 · Rank 1 / Foundation

Evaluate the definite integral $\int_0^2 x^3 dx$.

☐ A 4

☐ B 8

☐ C 16

☐ D 0

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Q2 · Rank 1 / Foundation

Evaluate the definite integral $\int_{-2}^2 x^3 dx$.

☐ A 8

☐ B 0

☐ C 4

☐ D -8

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Q3 · Rank 1 / Foundation

Find the exact area under the curve $y = (x - 2)^2$ between $x = 2$ and $x = 3$.

☐ A 1

☐ B 3

☐ C $\frac{1}{3}$

☐ D 0

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Q4 · Rank 2 / Core

Evaluate the integral $\int_1^3 4x^2 dx$.

☐ A 36

☐ B 108

☐ C $\frac{26}{3}$

☐ D $\frac{104}{3}$

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Q5 · Rank 2 / Core

Evaluate the integral $\int_1^2 (4x^3 - 3x^2 + 4x + 2) dx$.

☐ A 16

☐ B 20

☐ C 12

☐ D -16

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Q6 · Rank 2 / Core

Evaluate the integral $\int_{-1}^2 (5x^2 - 4x + 6) dx$.

☐ A $\frac{23}{3}$

☐ B 27

☐ C $\frac{52}{3}$

☐ D -27

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Q7 · Rank 3 / Application

Find the exact area under the curve $y = 2x^2 + x + 3$ between $x = 1$ and $x = 5$.

☐ A $\frac{320}{3}$

☐ B $\frac{334}{3}$

☐ C $\frac{248}{3}$

☐ D 53

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Q8 · Rank 3 / Application

Write down an expression for $g'(x)$ when $g(x) = \int_2^x (5t^2 - 2t) dt$.

☐ A $10x - 2$

☐ B $5x^2 - 2x - 16$

☐ C $5x^2 - 2x$

☐ D $\frac{5}{3}x^3 - x^2$

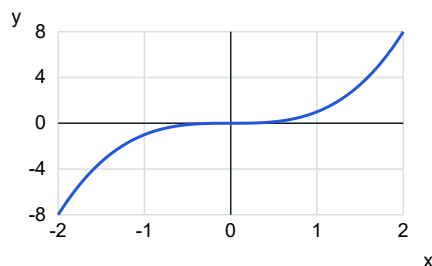
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Q9 · Rank 3 / Application

What is the actual area between the x axis and the graph of $y = x^3$, over the range $-2 \leq x \leq 2$?

☐ A 0☐ B 4☐ C -8☐ D 8

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Q10 · Rank 4 / Challenge · OPTIONAL

Evaluate the integral $\int_2^3 \frac{2}{x^3} dx$.

☐ A $-\frac{5}{36}$ ☐ B $\frac{1}{9}$ ☐ C $\frac{5}{36}$ ☐ D $\frac{19}{2}$

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